Unit - IV

Formal Models of Inductive generalization :-

⇒ Inductive generalization is the process of learning general rules (or) patterns from specific examples.

- In simple, it is the ability to :- - observe specific Inputs
  - Identify patterns.

Ex:- • Dog, cat, cow → all are animals.
→ Generalization: "These belong to the category animals".

- In CC, formal models use mathematical & computational methods to explain how this learning happens.

- formal models help to :- → Represent learning mathematically
  → Build intelligent sys that learn the data.

Types of formal models :-

(i) Rule based models :-
- Learn if - then rules from examples.

Ex :- "If an obj has wings" → It can fly.

used in :- - expert sys & Decision sys.

(ii) probabilistic Models :-

- uses probability to handle uncertainity.
  - It predict how likely a generalization is correct.

ex: • Bayesian Learning.

Advantage :- Handles uncertainity well.

(iii) Neural n/w models :-

- It learns patterns from data automatically.
- Generalize by adjusting weights.

ex: Image recognition sys.

Process of Inductive Generalization :-

① collect examples
② Identify patterns
③ form hypothesis (rule)
④ test & refine rule.

Importance :

- Enables Learning from data.
- Helps AI sys adapt to new situations.

Applications :-
- ML models
- pattern recognition
- NLP.

causality :-

- It refers to the relationship b/w cause & effect, where one event (cause), leads to another event (effect)

- In CC, causality helps systems understand why something happens, not just what happens.

ex :- • Smoking → causes lungs ~~cause~~ disease.
• Rain → causes wet roads.

Models of causality :-

(i) Causal Graphs (Bayesian N/w) :-

- Represents variables as nodes.
- Arrows show case - effect relations.

ex :- Rain → Traffic → Delay.

(ii) Structural equation Models (SEM) :-

- Mathematical Eqn represent casual relations.

(iii) counterfactual models :-

- Analyze "What if" scenarios.

ex :- "What if it had not rained?"

Types of Relationships:-

(i) Causal Relationship:-

→ Direct cause - effect link

ex:- Heat → ice melts.

(ii) Correlation:-

- Two variables related but not necessarily causal

ex:- Ice cream sales ↑ & temp ↑.

(iii) Spurious Relationship:-

- It is false (or) misleading relationships.

Importance:-

- It helps in deep understanding of problems.
- It Enables better predictions.
- It supports decision making.

Advantages:-

- Improves decision making
- Helps in prediction & control.

Limitations:-

- complex modeling
- Requires large & accurate data.

Applications:-

- Medical diagnosis
- AI decision sys.
- Recommendation sys.

categorization & Similarity :-

- These both are core cognitive processes used to organize knowledge & recognize patterns.

Categorization :-

- It is the process of Grouping obj based on common features (or) properties.

Ex :- • Dog, cat, cow → category : Animals.

Types :- (i) Rule based :- - These are based on fixed Rules.

ex :- "If an obj has wings → It is a bird".

(ii) prototype - Based :-

- It is based on an ideal example (prototype).

Ex :- "A "Sparrow" is a typical bird → used as reference.

(iii) Exemplar based :-

- It is based on stored examples.

ex :- "comparing a new animal with known animals.

Similarity :-

- It measures how closely two obj resemble each other.

Ex :- • Dog & wolf → high similarity

• Dog & car → low "

Types :-

(i) feature based :-

- compare common features.

(ii) Distance based :-

- Measure distance in feature space.

Less distance = more similarity.

(iii) Structural :-

- compare relationships b/w parts.

Relationship b/w Categorization & Similarity :-

- Similar obj → grouped into same category.
- categorization depends on similarity.

Ex :- New obj similar to "birds" → categorized as bird.

computational models :-

- clustering Algo
- Nearest Neighbor model.
- Neural N/w.

Importance :-

- Reduces complexity of info.
- Helps in faster decision making.
- enables pattern recognition.

Limitation :-

- categories may overlap.
- Difficult with complex data.

Applications :-

- Image recognition
- speech "
- NLP.
- pattern recognition

The role of analogy in problem solving :-

⇒ Anology is a cognitive process where we use knowledge from ~~simil~~ familiar situation to solve a new (or) unfamiliar problem (target).

- In CC, anology helps sys transfer knowledge & solve problems efficiently.

Ex:- • flow of water in pipes ⟷ flow of electricity in circuits.

Role of Analogy in problem solving :-

(i) Knowledge transfer :-

- Reuses past knowledge in new situations.

(ii) Simplifies complex problems :-

- Makes difficult problems easier by relating them to known ones.

(iii) Improves Learning:-

- Helps in understanding new concepts quickly.

(iv) Enhances creativity:-

- encourages innovative thinking & solutions.

Steps in Analogical problem solving:-

1. Retrieve a similar past problem (source)
2. Map similarities b/w source & target.
3. Transfer solution from source to target
4. Adapt the solution to ~~fi~~ fit the new problem.

Types of Anology:-

(i) Surface Analogy:-

- It is based on obvious similarities.

(ii) ~~Based on~~ Structural Analogy:-

- Based on deeper relationships.

- More imp for problem solving.

Computational models of Analogy:-

- Structure mapping model
- Case base Reasoning (CBR)
- Neural N/w models.

Advantages :- Reduces problem solving time.
- Improves understanding.

Limitations :-
- wrong anology leads to incorrect solutions.

Applications :-
- Education & learning sys
- scientific discovery.
- AI problem solving system.

Cognitive Development child concept acquisition :-

⇒ cognitive Development refers to how child's thinking, reasoning & understanding abilities grow over time.

- concept acquisition is the process by which children learn & form concepts (ideas, categories, meanings).

Ex :- A child Learns :
• Dog, cat, cow → all are "animals".

Stages of cognitive Development :- (Jean piaget)

- According to jean piaget, children develop concepts in stages :-

(i) Sensorimotor stage (0-2 years) :-
- Learning through senses & actions.
- Develops obj performance

ex :- knowing an obj exists even when hidden

(ii) preoperational Stage (2-7 years) :-

- It uses language & symbols.
- Thinking is intuitive, not logical.

ex :- Using words & images to represent obj.

(iii) concrete operational Stage (7-11 years) :-

- Logical thinking develops.
- understands categories & relationships.

ex :- Sorting obj. by size & shape.

(iv) formal operational Stage (11+ years).

- Abstract & logical thinking.
- problem solving ability improves.

Process of Concept Acquisition :-

(1) observation :- child observes obj/events.

(2) Comparision :- Identifies similarities & differences.

(3) Generalization :- forms general rules.
ex :- "All birds can fly".

(4) Refinement :- corrects mistakes
ex :- Learns penguins can't fly.

Importance :-

- Helps understand how humans learn.
- Improves AI learning models.

Applications :- → Educational technologies
→ child Learning sys.
→ AI - based learning models.
→ Intelligent tutoring sys.

Cognition & Artificial cognitive architectures such as ACT-R :-

⇒ Cognition refers to mental processes such as :- Thinking, memory, Learning & problem solving.

- In CC, Artificial cognitive architectures are computational models designed to stimulate human cognition.

- one of the most imp architectures is ACT-R.

ACT-R :- (Adaptive control of Thought - Rational).

⇒ It is a cognitive architecture developed by John R. Anders

- The Goal is to explain how humans think, Learn & Solve problems.

Components :-

① Declarative memory :- - Stores facts & knowledge.
ex :- "paris is capital of France".

② procedural memory :- - Stores skills & rules (Production rules).
- Ex! Steps to solve a problem.

③ Working ~~b~~ memory (Buffers) :-

- Temporary Storage for current tasks.

④ Modules :- ACT-R has different modules :-

- Visual module.
- Motor "
- Goal "

- These interact like human brain sys.

Working :-

① I/P is received.
② Info Stored in memory.
③ production rule are applied
④ Best rule is selected.
⑤ Action is performed.

Advantage :-

- Explains learning & memory
- Useful in AI sys design.

Limitations :-

- Complex to implement
- Requires detailed knowledge.

Applications :-

- Intelligent tutoring sys
- AI based Learning sys

SOAR :-

- It is an artificial cognitive architecture.
- It is used to model human intelligence & problem solving.
- It is designed to create sys that can learn, reason, & act like humans.
- It is developed by Allen Newell & his team.

Components :-

1. Working memory :-
   - It stores current info abt the problem.
2. Long term memory :-
   - Includes :- • procedural memory (rules)
     • semantic " (facts)
     • Episodic " (past experiences)
3. production Rules :-
   - Knowledge represented as if - then rules.

   ex :- IF condition → THEN action.

Working :-

1. observe current State
2. Apply rules to generate possible actions
3. select best action (operator)
4. execute action
5. Repeat until goal is reached.

Advantages :-

- General purpose Architecture.
- Supports learning & reasoning.

Limitations :-

- complex to design.

Application :-

- Robotics
- Intelligent agets.
- Decision support sys.

Opencog :-

- It stands for "open source artificial cognitive Architecture"
- It is designed to build Artificial General Intelligence (AGI).
- The Goal is to creates machines that can think, learn & reason like humans across different tasks.

Working:-

① I/p data is stored in Atomspace.
② Relationships are formed b/w concepts.
③ Reasoning engine processes knowledge.
④ Learning sys updates knowledge.
⑤ sys improves over time.

features :-

- It is open source architecture.
- It supports AGI development.
- It can Handle uncertainity & complex data.

Atomspace :- - It is core knowledge base.
- Stores info as a graph of nodes & links.

- It Represents :- • concepts
• Relationships & Actions.

Advantages :-
- flexible & scalable.
- combines different AI techniques.
- suitable for complex problems solving.

Limitations :-
- Still under development.
- complex architecture.

Applications :- - Robotics.
- NLP
- Intelligent agents.
- knowledge based sys.

Copy cat :-

- It is a cognitive architecture.
- It is designed to model human analogy-making & flexible thinking.
- It focuces on how humans:-
    - Recognize patterns
    - form analogies
    - solve problem creatively.

Ex:- "abc → abd"

Apply same rule to: "xyz → ?"

Answer :- "xyd"

(Based on anology : last letter changes)

Components :-

(i) workspace :- stores current problem & possible ~~soln~~ solutions.

(ii) Slipnet (concept N/w) :-

- N/w of concept (letters, positions, relations).

(iii) codelets :-

- It is small independent processes.
- performs simple tasks like:-
    - Detect patterns
    - Build structure
- work together to solve problems.

Working :-

① I/P problem is placed in workspace.

② Slipnet activates related concepts.

③ codelets explore possible patterns.

④ sys forms & test analogies.

⑤ Best solution emerges.

Advantages :-
- captures human like flexible thinking
- Handles ambigious problems.

Limitations :-
- works only in limited domains.
- complex to implement.

Applications:-
- pattern recognition
- Analogy based reasoning

Memory N/w :-

- These are AI models that can remember things & use them later.

- Normal neural n/w :- - learns patterns
  - But don't remember clearly.

- Memory N/w :- - store info (like memory).
  - uses it when needed.

Scanned with OKEN Scanner

Working :- ① Take i/p (question / data)

② Store it in memory

③ find imp information.

④ Use it to give answer.

Ex :- "Where is the book?"

memory :- "Book is on the table".

Answer :- "on table".

Main parts :-
- ~~Input~~ Input :- takes data
- Memory :- Stores info
- processing :- finds useful info
- o/p :- gives ans.

Advantages :-
- Remembers info
- Gives accurate answers
- works well with long data

Disad :-
- complex sys
- Needs more data

Applications :-
- chatbots
- Q/A sys
- AI assistants
- Language translation.